

## Objective

The objective of the Day 8 project is to develop a **Simple To-Do List application** using Python and Streamlit that allows users to manage their daily tasks efficiently. The application demonstrates the practical use of **Python lists, loops, conditional statements, functions, and menu-driven programming**. It implements basic CRUD operations, allowing users to add, view, update, and remove tasks through an interactive local web interface.

## Technologies Used

The application is developed using **Python** and **Streamlit**. Python lists are used to store tasks, while functions are created for different operations such as adding, updating, and removing tasks. Loops are used to display multiple tasks, and conditional statements are used for decision-making and input validation. Streamlit provides the interactive web interface and `st.session_state` is used to maintain the task list during user interaction.

## Working Principle

The application starts with an empty task list. The user can select an operation from the menu, such as adding, viewing, updating, or removing a task. When a new task is added, it is stored in the Python list. The view operation displays all stored tasks with their corresponding numbers. For updating a task, the user selects its number and enters the new task description, which replaces the existing task. The remove operation deletes the selected task from the list. The application also checks for invalid or empty inputs and displays appropriate messages to the user.

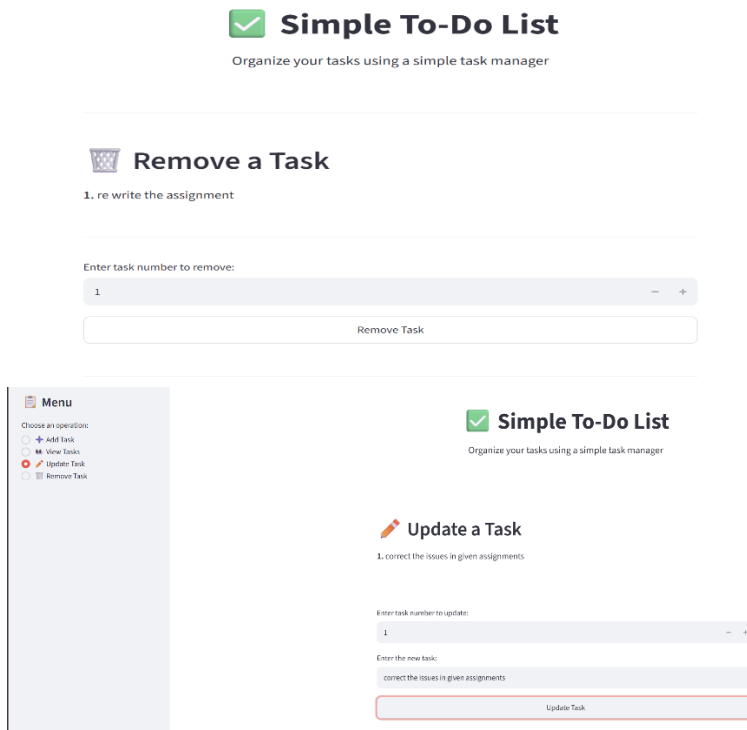
## Learning Outcomes

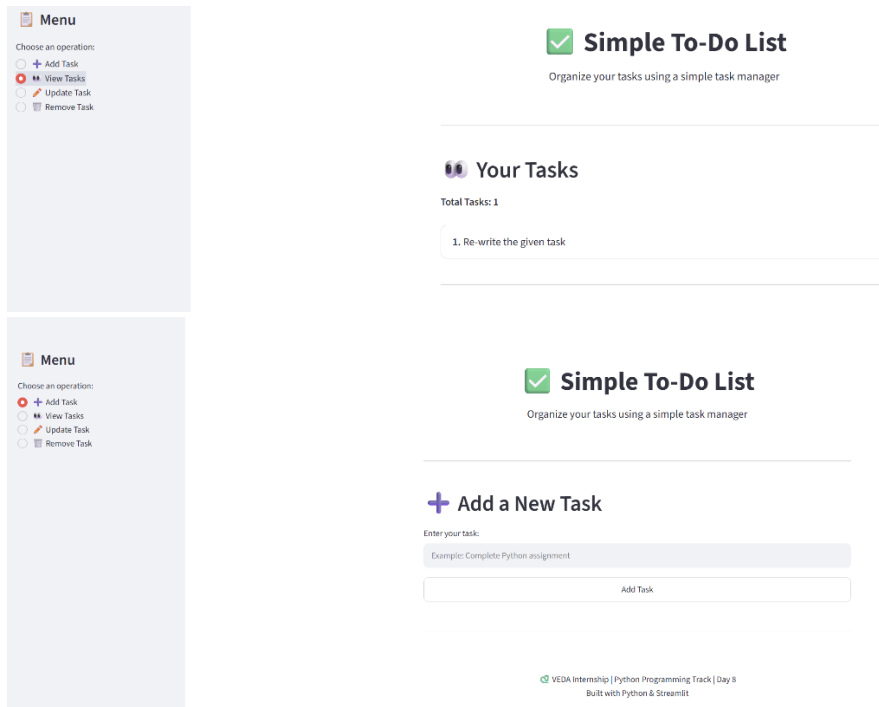
This project provides practical experience in developing a **menu-driven application** and understanding how CRUD operations work. It improves knowledge of Python lists, functions, loops, conditional statements, indexing, input validation, and Streamlit session state. The project also demonstrates how basic Python programming concepts can be combined to create a useful task-management application.

C:\Users\shamb\OneDrive\Desktop\VEDA internship\DAY8>

```
app.py > ...
1 import streamlit as st
2
3 st.set_page_config(
4     page_title="Simple To-Do List",
5     page_icon="✅",
6     layout="centered"
7 )
8
9 if "tasks" not in st.session_state:
10     st.session_state.tasks = []
11
12 def add_task(task):
13     """Add a new task to the list."""
14     if task.strip():
15         st.session_state.tasks.append(task.strip())
16         return True
17     return False
18
19 def update_task(task_number, new_task):
20     """Update an existing task."""
21     index = task_number - 1
22
23     if 0 <= index < len(st.session_state.tasks):
24         if new_task.strip():
25             st.session_state.tasks[index] = new_task.strip()
26             return True
27         return False
28
29 def remove_task(task_number):
30     """Remove a task from the list."""
31     index = task_number - 1
32
33     if 0 <= index < len(st.session_state.tasks):
34         st.session_state.tasks.pop(index)
35         return True
36     return False
37
38 st.markdown("""
39 <style>
40 .title {
41     text-align: center;
42     font-size: 42px;
43     font-weight: bold;
44     margin-bottom: 5px;
45 }
46
47 .subtitle {
48     text-align: center;
49     font-size: 18px;
50     margin-bottom: 30px;
51 }
52
53 .task-card {
54     padding: 12px 18px;
55     margin: 8px 0;
56     border-radius: 10px;
57     border: 1px solid #ddd;
58     font-size: 17px;
59 }
60
61 .footer {
62     text-align: center;
63     margin-top: 35px;
64     font-size: 14px;
65 }
66 </style>
67 """, unsafe_allow_html=True)
```

<http://localhost:8501>





## Simple To-Do List

Organize your tasks using a simple task manager

### Remove a Task

 No tasks available to remove.

<https://github.com/shambhavi0804/VEDA - internship/commit/61eacfbdd5f5738c4bcd0a3b8ddb870750b3cb3>

```
1  @ - 5,8 +1,274 @@
2  + import streamlit as st
3  +
4  + # Page Configuration
5  + #
6  + st.set_page_config(
7  +     page_title="Simple To-Do List",
8  +     page_icon="📌",
9  +     layout="centered"
10 + )
11 +
12 + #
13 + # Initialize Task List
14 + #
15 + if "tasks" not in st.session_state:
16 +     st.session_state.tasks = []
17 +
18 +
19 + #
20 + # CRUD Functions
21 + #
22 +
23 + def add_task(task):
24 +     """Add a new task to the list."""
25 +     if task.strip():
26 +         st.session_state.tasks.append(task.strip())
27 +         return True
28 +     return False
29 +
30 +
31 + def update_task(task_number, new_task):
32 +     """Update an existing task."""
33 +     index = task_number - 1
34 +
35 +     if 0 <= index < len(st.session_state.tasks):
36 +         if new_task.strip():
37 +             st.session_state.tasks[index] = new_task.strip()
38 +             return True
39 +         return False
40 +     return False
```

<https://lnkd.in/p/dzxsXDXu>

